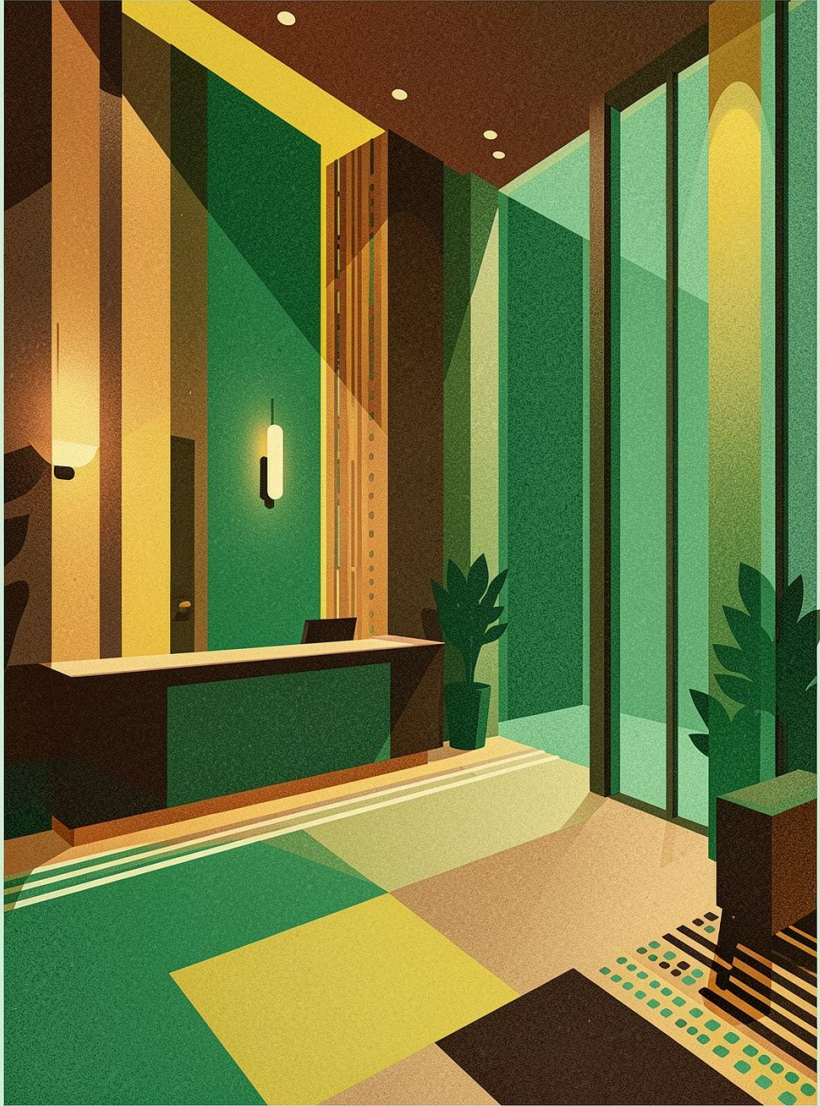




Hotel Revenue Operations

Predicting Cancellation Risk & Protecting Revenue with Machine Learning

Laxmi Gupte | Data Scientist

1 in 4 Hotel Bookings Gets Cancelled

Hotels lose revenue, waste staff time, and can't plan occupancy accurately when bookings cancel at the last minute. Most teams only find out when it's too late to act.

27.5%

Cancellation Rate

Across 87,396 real bookings

€11.5M

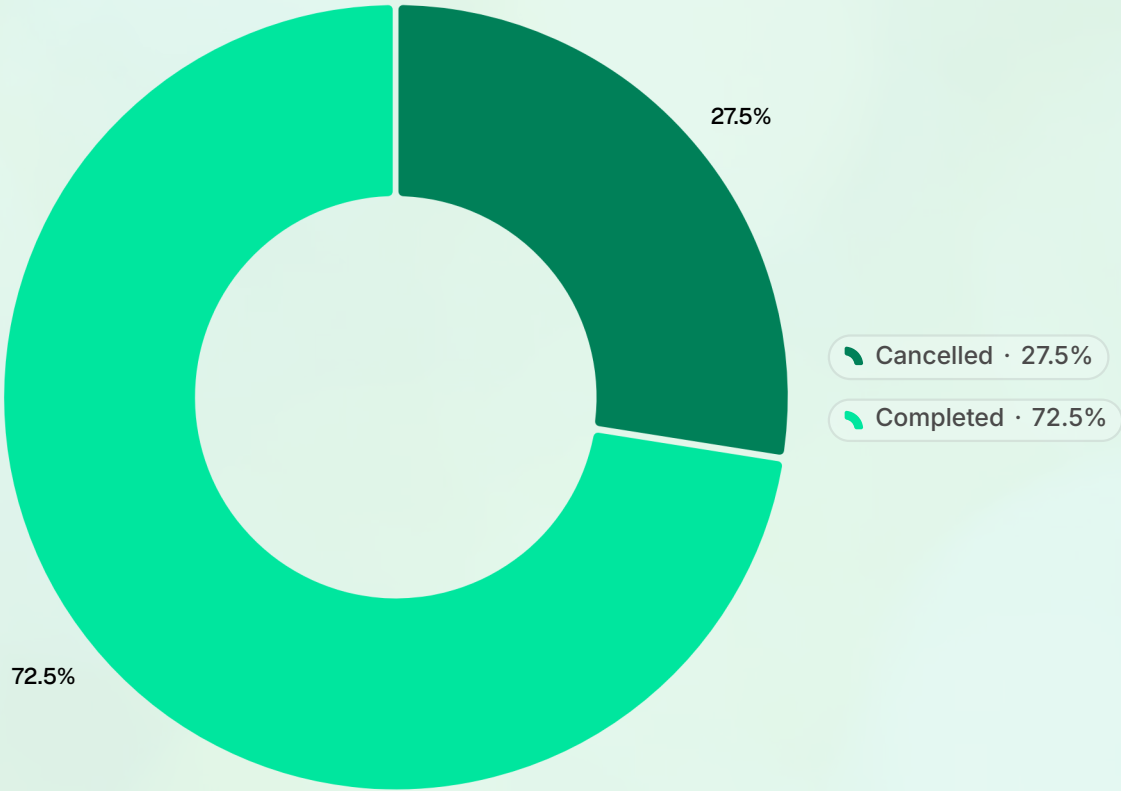
Revenue at Risk

Total exposure across the portfolio

3 yrs

Data Span

2 hotels · Portugal · 2015–2017



Nearly **24,000 bookings** cancelled — each one a missed revenue opportunity and a planning failure the team couldn't predict in time.

Build a System That Spots Risky Bookings Before It's Too Late

Three goals drove every decision in this project. The output isn't a research paper — it's a working tool hotel managers can open on Monday morning.

Predict

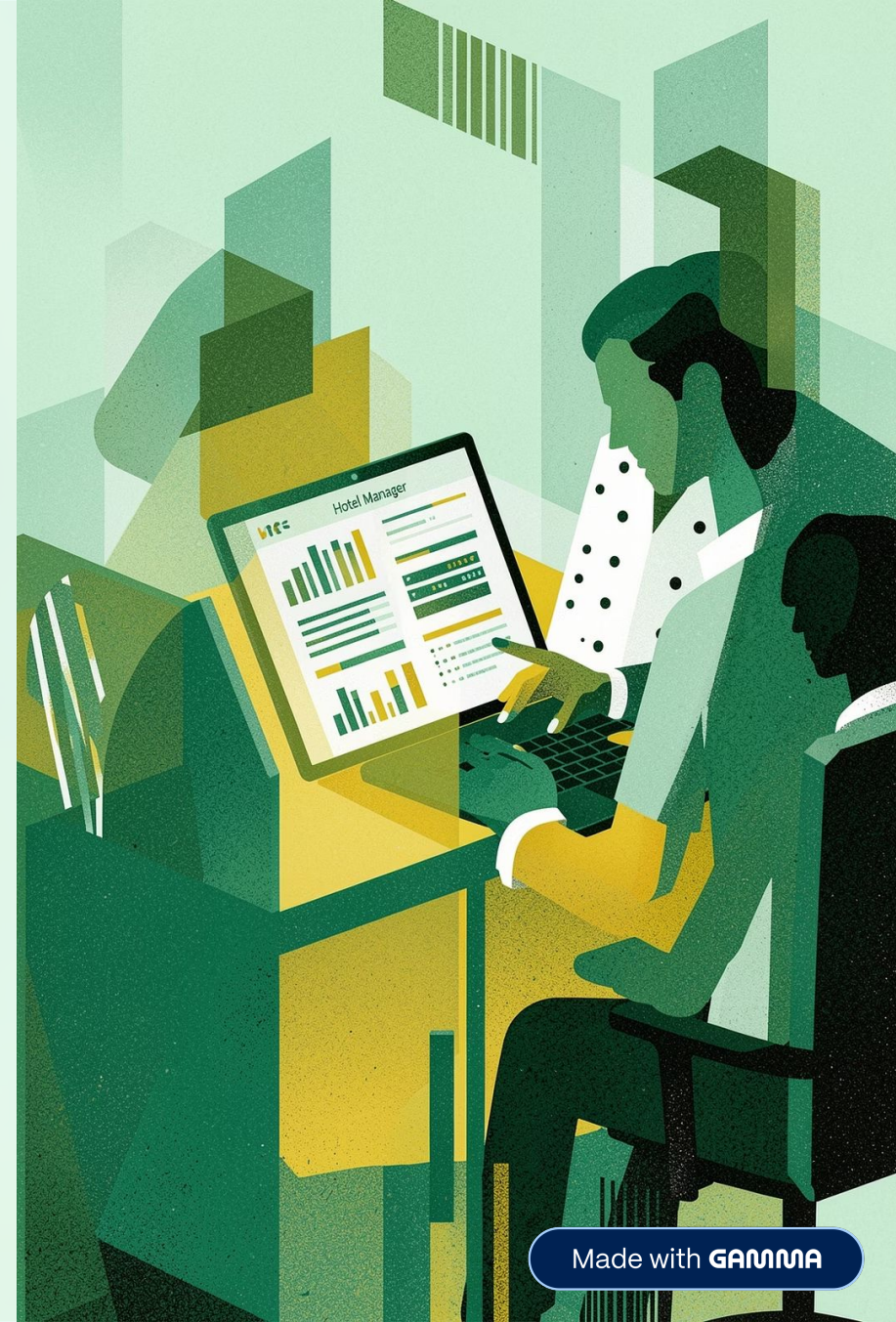
Which bookings are most likely to cancel — scored in real time, for every new reservation.

Explain

Why is this booking high risk? What signals drove the score? Plain-English reasons, not black-box outputs.

Act

Which bookings should the revenue team prioritise today? Turn predictions into a daily action list.



87,396 Real Hotel Bookings Across 2 Properties

Every row is a real guest interaction. This is the same type of data any hotel already collects in their property management system.



City Hotel + Resort Hotel

Two distinct property types with two different cancellation profiles — capturing the full range of guest behaviour.



Portugal · 2015–2017

Three full years of seasonal variation — peak summer, off-season, and everything in between.



32 Variables per Booking

Channel, lead time, pricing, deposit type, guest history, room type, and special requests — the full picture of a reservation.

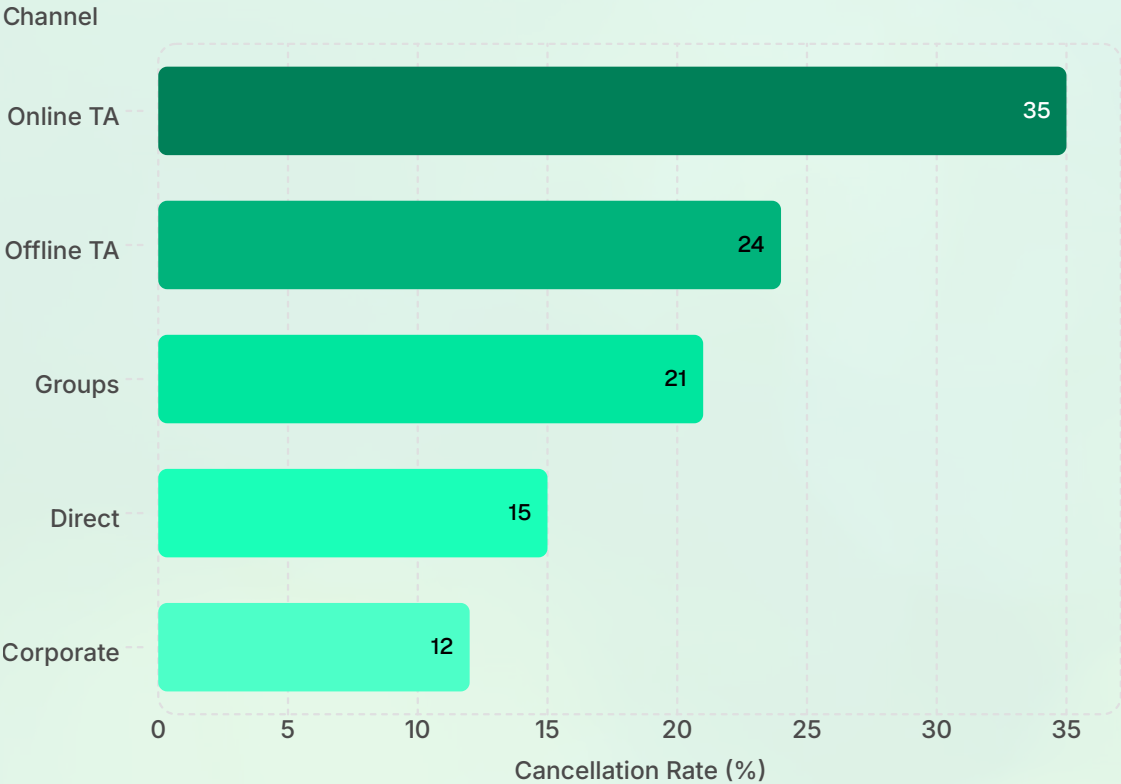


No Synthetic Data

Every insight is grounded in real guest decisions — not simulated scenarios or augmented records.

Not All Bookings Are Equal — Some Are Far Riskier Than Others

Four patterns emerged clearly from the data *before* any model was built. Risk is predictable — and it clusters around specific booking behaviours.



Lead Time

180+ days out: **40%** cancel rate. Last-minute bookings: just **16%**.

Deposit Behaviour

99% of reservations carry no deposit — the lowest-commitment guests dominate the portfolio.

Guest Engagement

Zero special requests → **35%** cancel rate. Two or more requests → just **15%**.

The Model Reads Each Booking Like a Risk Checklist

For every new booking, the system evaluates 41 signals and outputs a single number: the probability this booking will cancel. Think of it as a credit score — but for hotel reservations.

Low Risk

Repeat guest · Direct booking · 5-day lead time

2%

Cancel probability

Medium Risk

Online TA · 90-day lead time · No deposit

38%

Cancel probability

High Risk

Online TA · 200-day lead · No deposit · Prior cancellation

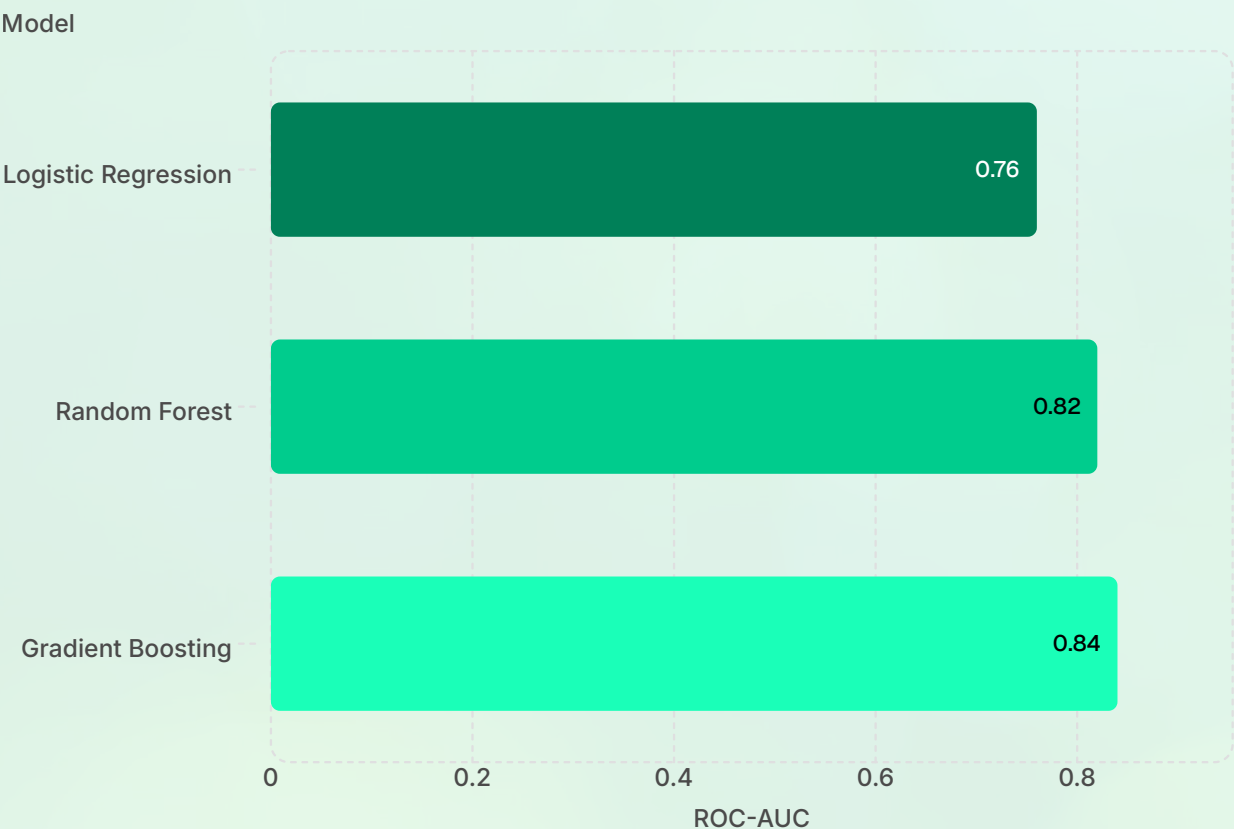
99%

Cancel probability

 Each score is accompanied by a plain-English explanation — so the revenue manager knows **why** a booking is flagged, not just **that** it is.

The Model Correctly Identifies 8 in 10 Cancellations

Four models were tested head-to-head. Gradient Boosting won — it learns from patterns across thousands of past bookings to score new ones in real time.



ROC-AUC = how well the model separates risky bookings from safe ones. A score of 1.0 is perfect; 0.5 is a coin flip. At **0.84**, the model is meaningfully better than chance — and better than all alternatives tested.

0.84

ROC-AUC Score

Gradient Boosting — winning model

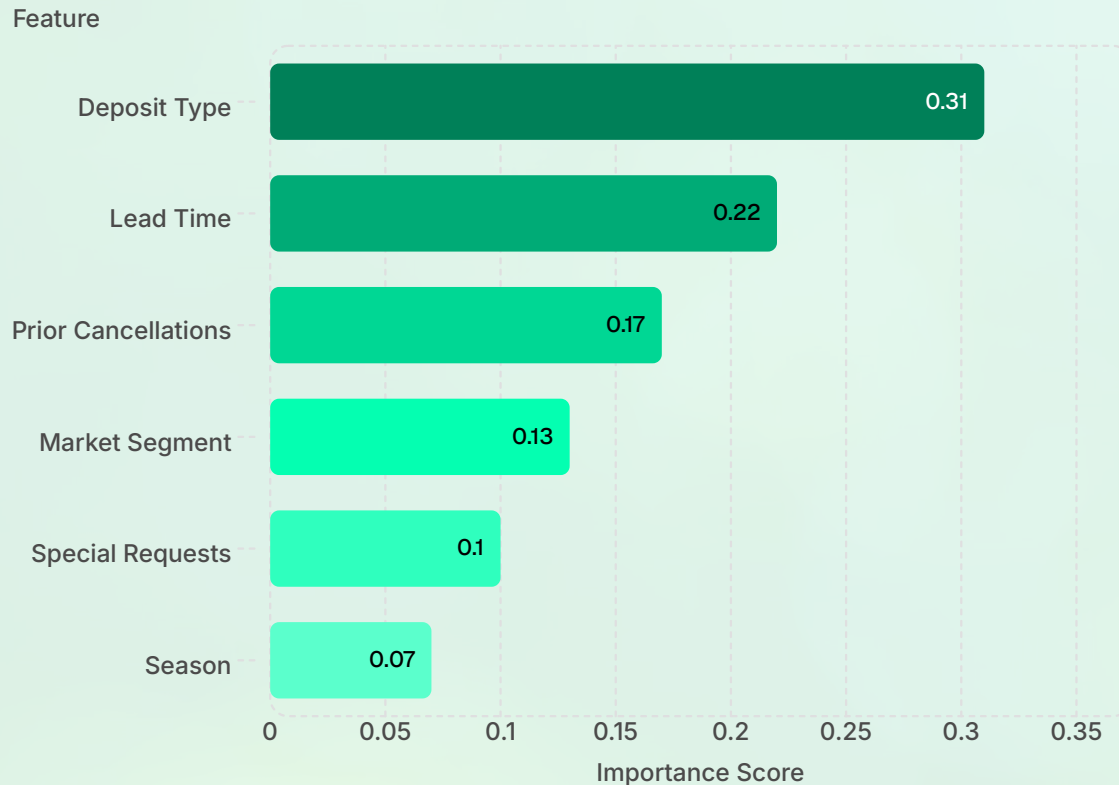
0.26

Decision Threshold

Set to catch more real cancellations — a missed cancellation costs more than a wasted phone call

The Biggest Risk Signals — In Order

These are the factors the model weighted most heavily when predicting cancellation. Each one maps directly to a business decision a hotel manager already understands.



→ Deposit Type (0.31)

No financial commitment = higher risk of walking away

→ Lead Time (0.22)

The longer the wait, the more likely plans change

→ Prior Cancellations (0.17)

Past behaviour predicts future behaviour

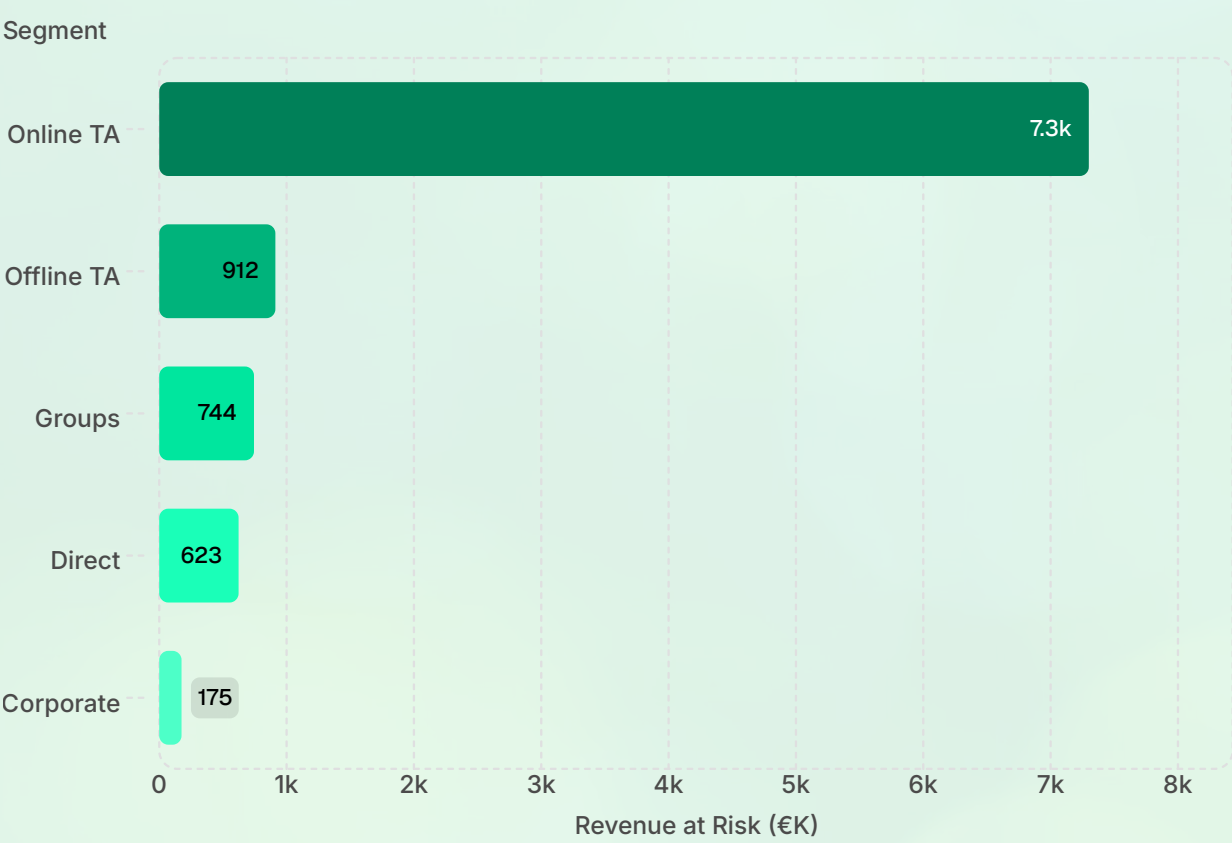
→ Special Requests (0.10)

Engaged guests are invested guests — less likely to cancel

❏ **Intentional exclusion:** Room-type assignment data was deliberately left out — it's only known after check-in, which would give the model an unfair advantage on historical data. This is a data integrity decision, not an oversight.

Online TA Accounts for the Largest Share of Revenue at Risk

This tells hotel teams not just *who* is likely to cancel — but which cancellations will hurt the most financially. OTA risk dwarfs every other segment combined.



€7.3M

Online TA at Risk

The single largest exposure — by a wide margin

€11.5M

Total Portfolio Risk

Across all channels and both properties

OTA represents **63%** of total revenue at risk despite being one of several booking channels. The model's output directly informs where to concentrate retention effort.

The Model Lives Inside a Tool Hotel Managers Can Actually Use

Built in Streamlit — a web app that runs in any browser, no technical setup required.
Dark UI with teal and coral accents. Designed for non-technical users.



Manual Prediction

Enter a new booking's details → instant risk score + plain-English explanation of what drove it



Explain a Booking

Upload an existing booking → see exactly which factors pushed the risk up or down



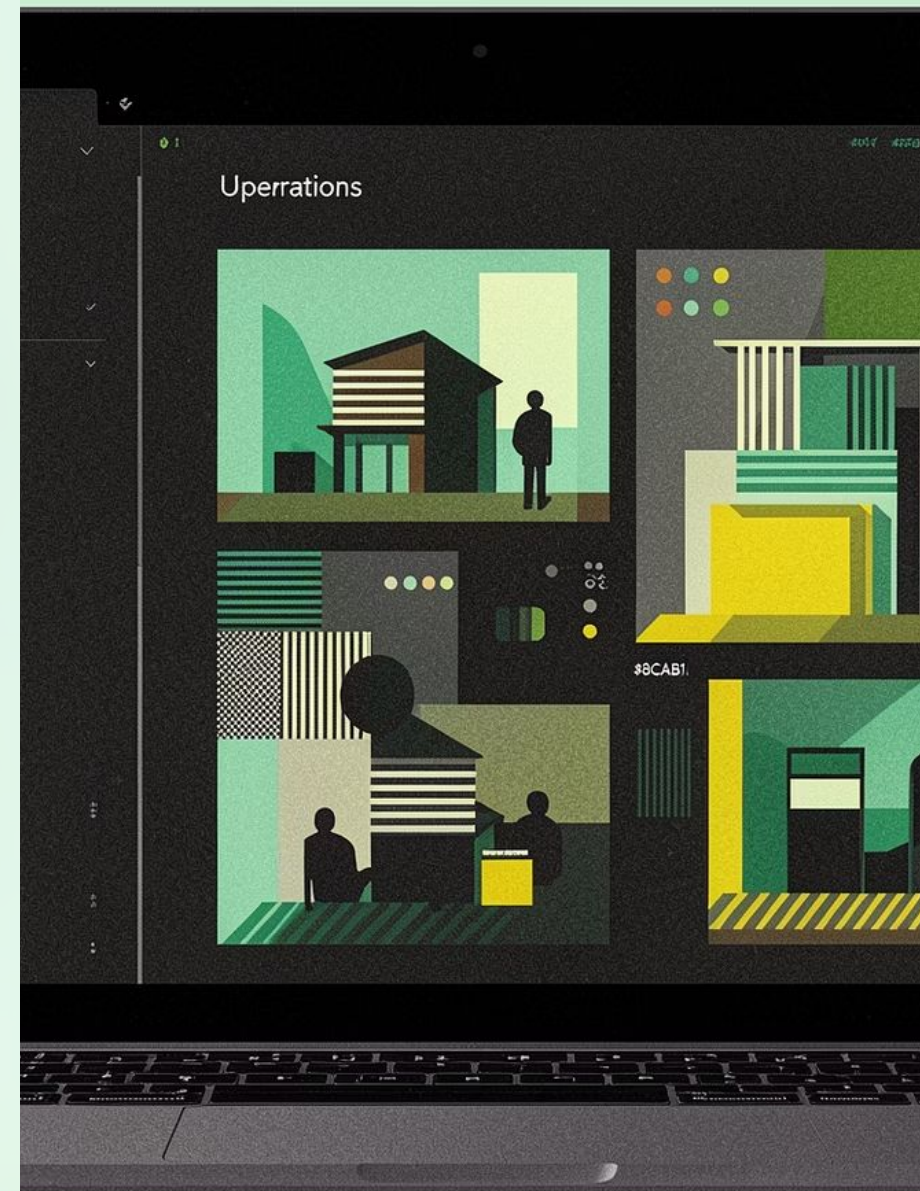
Batch Scoring

Upload a full CSV of bookings → score all of them at once, revenue-at-risk flagged per row



Segment Dashboard

Portfolio-level view of cancellation risk by hotel, channel, and season — for weekly revenue reviews



What Was Built, What Was Fixed, and What Was Left Out Intentionally

Documenting what was wrong and fixing it openly is a sign of engineering maturity — not a weakness.

✓ Built

- End-to-end ML pipeline
- Cost-weighted decision threshold
- Shared feature engineering module
- SQL analytical layer
- Automated test suite
- CI/CD via GitHub Actions

🔧 Fixed from Original

- Silent encoding bug that broke 2 key features
- App/model vocabulary mismatch
- Notebooks that crashed on fresh run
- Missing children null values

🚫 Left Out on Purpose

- Room-type assignment data — leakage risk
- XGBoost/SHAP — environment constraints, but architecture transfers directly

📄 Intentional exclusions are documented decisions — each one has a defensible reason tied to data integrity or deployment constraints.

